

..Weaker force

Farther away from the poles, the magnetic force grows weaker. Here the iron filings are not pulled into lines.

If you cut a magnet in half, you would get two magnets, each with its own north pole and south pole.

.....South pole

This end of the magnet is the south pole. If it meets a south pole on another magnet, it pushes it away. If it meets a north pole on another magnet, it pulls towards it.

.....Stronger force

Lots of iron filings are pulled toward the poles of the magnet because that is where the magnetic force is at its strongest.

Magnets at work

A compass needle is a magnet. One end always points north because the Earth itself acts as a giant magnet—it has a north pole and a south pole. The Earth's magnetic field turns the needle until it follows the Earth's lines of magnetic force.



Compass

between the poles.